From Prediction to Conditional Exploration in LLM Agent Simulations for AI Policy Governance

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Abstract

Large Language Model (LLM)–based agent simulations are increasingly proposed as tools for informing AI governance. However, policymaking is inherently uncertain, high-stakes, and socially embedded. This position paper argues that LLM-agent simulations should not function as predictive policy engines but as structured uncertainty exploration systems. Drawing on generative agent architectures and recent multi-agent simulation work, I examine two challenges: interpreting conditional simulation outcomes under deep uncertainty, and establishing transparency in architecturally opaque systems. I propose a co-evolutionary framework integrating robustness stress-testing and layered auditability, reframing simulation from prediction toward conditional exploration and institutional accountability in AI policy design.

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

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1 Introduction

Policy decisions are rarely impulsive. They are typically negotiated, evidence-informed, and shaped through institutional deliberation, stakeholder consultation, and political accountability. With the advancement of technology, it has come to a phase where the process somewhat automates using AI agent simulations [9]. I begin from this premise: policymaking is thoughtful, iterative, and constrained by real-world consequences. As such, any claim that LLM-based agent simulations can directly "generate" policy must be approached with caution.

Recent advances in generative agents demonstrate that Large Language Models can simulate believable individual and collective behaviors through architectures that integrate memory, reflection, and planning [12]. These systems exhibit emergent properties such as information diffusion, opinion formation, and coordination. More recently, multi-agent simulations such as VACSIM have explored whether LLM-powered societies can approximate real-world public health dynamics and inform intervention strategies [7]. These suggest that generative simulations may function as in silico testbeds—spaces to rehearse possible futures before enacting policy in the real world.

I agree with the emerging argument that the value of LLM agent simulations does not emerge "out of the box." Rather, it emerges through iterative, stakeholder-engaged design. However, I argue that we must go further. The central challenge is not simply building believable agents; it is understanding how to simulate complex reactions to AI systems themselves—particularly when AI technologies are the object of policy intervention.

Consider disinformation generated by AI systems. If a government proposes a policy to regulate AI-generated content, what exactly should be simulated? The spread of synthetic media? The adaptive responses of malicious actors? Public trust erosion? Platform moderation feedback loops? The behavioral architectures described in generative agents[12], and the demographic and network modeling approaches in VACSIM [7] provide starting points for modeling diffusion and attitude change. Yet AI-driven disinformation ecosystems are reflexive: actors respond to policy, platforms update algorithms, and generative models themselves evolve. Simulating such dynamics requires modeling not only social interaction, but also algorithmic mediation and adversarial adaptation.

This leads me to two core questions for policy generation under uncertainty:

(1) When simulations operate under probabilistic assumptions and partial representations of reality, how should policymakers interpret simulated outcomes in uncertain, high-stakes contexts?

If a simulated society suggests that a stricter moderation policy reduces misinformation spread, is that result robust across demographic subgroups, network structures, and model temperatures? As VACSIM demonstrates, even alignment with real-world trends requires parameter modulation and warm-up calibration [7]. Outcomes depend on modeling choices. Thus, simulated policy "effects" must be treated as contingent explorations rather than predictive certainties.

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(2) How do we establish transparency and accountability in simulations that themselves rely on opaque AI systems?

Generative agents rely on memory retrieval heuristics, importance scoring, and reflection mechanisms [12] where VACSIM introduces attitude modulation to correct for training biases [7].

These architectural decisions shape outcomes. If such simulations are used to recommend AI regulation, then the epistemic foundations of the simulation—prompt design, demographic sampling, network construction, temperature scaling—must be auditable. Otherwise, we risk replacing one opaque AI system with another opaque meta-system used for policymaking. Furthermore, AI algorithmic risks themselves—bias amplification, content personalization, feedback loops, and economic incentives—must be explicitly represented within simulations. Disinformation harms do not arise solely from user behavior; they emerge from socio-technical entanglements between algorithms, platforms, and publics. Simulating policy impact therefore requires modeling algorithmic mediation as an active participant in the system, not merely as a background variable.

I therefore position LLM agent simulations not as policy engines, but as structured uncertainty exploration tools. Their value lies in making assumptions visible, stress-testing policy narratives, and identifying unintended consequences before deployment. To achieve this, simulations must be co-developed with policymakers, domain experts, and affected communities, integrating participatory design principles from HCI and rigorous validation and verification standards from agent-based modeling [7].

In this position paper, I argue that responsible policy-oriented simulation must:

- Treat simulated outcomes as conditional hypotheses rather than prescriptions;
- Embed transparency mechanisms that surface architectural and parameter choices;
- Explicitly model algorithmic risks and adaptive adversarial behavior;
- Align simulation development with policy deliberation processes through iterative, stakeholder-engaged design.

Only through this co-evolutionary approach can LLM agent simulations move beyond technical novelty and become credible tools for navigating AI governance under uncertainty.

2 Related work

2.1 Interpreting Simulation Outputs Under Deep Uncertainty in Probabilistic Models and Partial Worlds

A longstanding concern in modeling scholarship is that simulations necessarily operate under probabilistic assumptions and partial representations of reality. Oreskes et al. [11] argue that models cannot be "validated" in any absolute sense; they can only be confirmed relative to specific questions and assumptions. Under this epistemological framing, simulated outcomes must be treated as conditional on model structure rather than as predictive truths.

In policy analysis, Decision Making under Deep Uncertainty (DMDU) and Robust Decision Making (RDM) extend this position [15] by arguing that when systems are complex and adaptive, computational models should be used to stress-test policies across plausible futures, not to forecast single outcomes [4]. This distinction is crucial for AI governance contexts such as misinformation regulation, where both adversaries and platforms adapt dynamically.

Agent-based modeling (ABM) research further emphasizes that outcomes in complex systems are highly sensitive to structural assumptions, including network topology, update rules, and parameter values [16]. Sensitivity analysis is therefore not an optional add-on but a core methodological requirement for credible policy interpretation.

The generative-agent literature reinforces these concerns. Generative Agents introduce memory streams, reflection modules, and planning mechanisms to simulate believable social behavior. However, emergent dynamics depend on retrieval heuristics (relevance, recency, importance), and failures often arise from improper memory retrieval or fabrication [12] . Thus, collective behaviors in such systems are shaped not only by "social dynamics" but also by architectural design.

Similarly, VACSIM explicitly frames simulation epistemology in terms of verification and validation [12]. Importantly, the system requires simulation warm-up stages and parameter modulation to avoid drastic early shifts and align initial attitudes with real-world baselines. These calibration choices illustrate that apparent policy effects depend on modeling decisions.

2.2 Robustness Across Subgroups and Structures

Policy interpretation becomes particularly fragile when simulations suggest that a stricter moderation policy reduces misinformation spread. The key scholarly question is: Is the effect robust? In ABM and computational social science, robustness requires exploring variation across demographic subgroups, social network structures, and behavioral rules. Small changes in network density or clustering can radically alter diffusion dynamics [14]. Likewise, parameter sensitivity studies show that policy conclusions can reverse under alternative calibration settings [3].VACSIM operationalizes this concern by averaging across multiple random seeds and analyzing policy effect differences under varying effort levels [7]. It also introduces attitude modulation via temperature scaling to address model bias. These adjustments demonstrate that even alignment with real-world trends requires deliberate calibration.

From a DMDU perspective, this implies that simulated policy "effects" should be interpreted as contingent explorations rather than predictive certainties. In high-stakes AI governance—such as regulating generative misinformation—models must reveal the boundaries of their assumptions. Rather than asking whether a policy works in one simulated world, policymakers must ask: Under what structural conditions does it fail? The literature therefore converges on three interpretive principles: 1) Treat simulated outcomes as conditional on structural and probabilistic assumptions. 2) Require systematic robustness analysis across parameters and subgroup structures. and 3) Frame simulations as exploratory scenario tools rather than forecasting engines.

2.3 Architectural Opacity in LLM-Based Agents Transparency and Accountability

The second question concerns simulations that rely on opaque AI systems. Generative agents depend on architectural mechanisms—memory retrieval scoring, reflection synthesis, importance weighting, and planning recursion [12]. These mechanisms directly shape behavior trajectories. When retrieval fails or hallucination occurs, emergent social patterns may reflect artifacts rather than plausible human dynamics [12]. VACSIM introduces additional architectural interventions, such as attitude modulation to counteract pre-training biases [7]. While such techniques improve alignment, they also highlight the constructed nature of simulation outcomes. Prompt design, demographic sampling, social network generation, and temperature scaling all influence final policy recommendations. Thus, when simulations are used to inform AI regulation, the epistemic foundations of the system must be visible.

Algorithmic accountability scholarship provides relevant guidance. Kroll et al. [8] argue that accountability can be achieved through structured oversight and auditable processes even when full model transparency is infeasible. Diakopoulos [5] emphasizes the need for mechanisms that allow stakeholders to interrogate decision systems. In AI documentation practices, Model Cards [10] and Datasheets for Datasets [6] propose standardized reporting of intended use, subgroup performance, data provenance, and limitations. In NLP, Data Statements (Bender & Friedman, 2018) highlight the importance of explicitly documenting demographic and linguistic coverage to prevent hidden biases [2].Applied to policy-oriented LLM simulations, transparency must extend beyond the base model to include: Prompt templates and policy framing language, Persona construction and demographic distributions, Network topology assumptions, Memory retrieval formulas and importance weights, Temperature scaling and calibration procedures and Random seed selection and aggregation methods. Without such documentation and auditability, there is a risk of replacing one opaque AI system (e.g., a content moderation algorithm) with another opaque meta-system used to justify regulation.

2.4 From Explainability to Institutional Accountability

Transparency literature distinguishes between explanation and accountability. Providing narrative summaries of agent behavior is insufficient. Sandvig et al. [13] emphasize algorithm auditing as a method for detecting structural biases. True transparency requires reproducibility, traceability, and the possibility of external review.

VACSIM gestures toward this direction by conducting multiple-run averaging and subgroup analyses [7] , and by evaluating consistency at both global and local levels. However, if such simulations are deployed for AI governance, stronger mechanisms would be required: parameter registries, audit logs, sensitivity reports, and explicit uncertainty communication. Across both questions, scholarship converges on a shared principle: LLM-agent simulations are not policy oracles but structured uncertainty exploration systems.

For uncertain, high-stakes contexts, interpretation requires robustness analysis, scenario exploration, and acknowledgment of contingency. For transparency, legitimacy requires documentation, auditability, and institutional oversight mechanisms that expose how architectural and parameter choices shape outcomes. If these conditions are not met, simulations risk becoming persuasive but epistemically fragile tools—producing the appearance of evidence while embedding opaque assumptions. The literature therefore suggests that responsible policy-oriented simulation must be co-developed with robust uncertainty protocols and transparent governance structures [1].

3 Conceptual Framework: Co-Evolving Policy and Simulation Under Uncertainty

I propose a framework that treats LLM-based agent simulations not as predictive engines, but as iterative policy co-design infrastructures. The framework is grounded in two core commitments derived from prior scholarship and the limitations surfaced in generative agent systems [7] and [12].

(1) Simulation outputs are contingent on modeling assumptions.

(2) Simulation architectures are themselves algorithmic systems requiring governance.

This section introduces a structured framework that connects these commitments to methodological practice.

3.1 Conditional Policy Effects, Not Predictions

When simulations operate under probabilistic assumptions and partial representations of reality, their outputs must be interpreted as conditional claims. Generative agents rely on memory retrieval heuristics, recency and importance scoring, and reflective synthesis [12].These components are not merely implementation details; they shape how agents remember information, update beliefs, and respond to policy interventions. VACSIM further demonstrates that simulated population dynamics depend on calibration procedures such as warm-up stages and temperature-based attitude modulation [7]. Within this context, a simulated result—for example, that stricter moderation reduces misinformation spread—cannot be interpreted as a direct forecast of real-world impact. It is instead a conditional outcome under specified assumptions about network structure, demographic distribution, memory decay, retrieval weighting, and sampling randomness.

This reframing shifts the epistemic role of simulation. Rather than asking "Does the policy work?", the relevant question becomes: " Under what modeling assumptions does the policy appear effective, and how stable is that appearance?" This conditional perspective aligns simulations with traditions of robust decision-making under deep uncertainty. The aim is not to eliminate uncertainty, but to expose its contours.

3.2 Robustness as a First-Class Design Requirement

To make conditional reasoning operational, robustness must become central rather than peripheral. Policy simulations should not present single-run outcomes or average trajectories alone. Instead, they should probe how results vary across structural and algorithmic perturbations.

Structural robustness concerns how outcomes change under alternative network topologies, information recommender dynamics, or exposure patterns. In misinformation governance contexts, diffusion depends heavily on clustering, bridging nodes, and algorithmic amplification effects. Small structural shifts can generate qualitatively different cascades.

Demographic robustness examines subgroup variation. VACSIM's design highlights how simulated agents are instantiated using demographic distributions [7], yet differential responses may emerge depending on how those demographics are parameterized. If stricter moderation reduces misinformation exposure overall but disproportionately suppresses certain community narratives, aggregate improvement may mask inequity. Algorithmic robustness interrogates the sensitivity of results to modeling parameters such as temperature scaling, retrieval thresholds, and decay rates [7].If policy conclusions invert under minor parameter adjustments, then the claim is fragile.

In this framework, robustness analysis becomes a methodological requirement rather than a supplementary evaluation. Policy recommendations must demonstrate bounded stability across structured perturbations. Where instability emerges, simulations should report fragility zones explicitly.

3.3 The Simulation Transparency Stack

If simulations are to inform AI regulation, they must not replicate the opacity of the systems they aim to govern. Generative agent architectures involve multiple layers of decision logic: memory scoring, reflective summarization, planning decomposition, and natural language prompting [12]. VACSIM adds demographic sampling, social network generation, and temperature modulation. Each layer embeds assumptions that shape outcomes. I therefore conceptualize transparency as a layered stack: At the representational level, simulations must document how agents are instantiated: what populations are included, what behavioral capacities are modeled, and what contexts are excluded. At the architectural level, memory retrieval formulas, reflection prompts, and planning logic must be exposed. Retrieval errors and fabricated embellishments observed in generative agents illustrate how unseen architectural behavior can distort emergent patterns. At the parameter level, temperature settings, decay factors, calibration procedures, and random seed strategies must be logged and reproducible. At the policy encoding level, simulations must disclose how regulatory interventions are translated into prompt language and behavioral constraints. Policy framing influences agent interpretation and downstream effects. Finally, at the aggregation level, simulations must report variance, subgroup breakdowns, and sensitivity ranges rather than only mean trajectories. Without such layered transparency, simulation-driven policymaking risks becoming an opaque meta-system—one algorithmic layer removed from the AI systems under regulation.

3.4 Reflexive Co-Development Between Policy and Simulation

The final element of this framework is reflexivity. Simulations and policies should not be sequential—where simulations produce answers and policymakers implement them—but iterative. In this co-development model, policymakers first articulate a regulatory hypothesis, such as mandating watermarking of AI-generated content. Modelers then translate that hypothesis into explicit structural and behavioral assumptions. Simulation outputs are generated under multiple perturbations. Instead of receiving a binary recommendation, policymakers receive a stability map: where the intervention reduces spread, where it fails, and where unintended consequences arise. This information feeds back into policy design. Policymakers may narrow scope, adjust enforcement mechanisms, or introduce safeguards based on identified fragilities. Simulation architecture is subsequently refined to capture newly surfaced dynamics. Through this loop, simulations become deliberative tools rather than deterministic advisors.

4 Discussion and Conclusion

This position paper advances an operational pathway for integrating LLM-agent simulations into AI governance without overstating their epistemic authority. The proposed methodology centers on extending generative agent systems to better reflect misinformation ecosystems, including algorithmic amplification and adversarial adaptation, and subjecting them to structured robustness testing across structural, demographic, and algorithmic dimensions. Rather than presenting single effect sizes, simulations should report stability ranges and fragility zones, explicitly identifying where policy conclusions shift under parameter or modeling changes.

Another pillar of the framework is simulation auditability. Given that generative agents rely on memory scoring and reflective mechanisms and that multi-agent simulations often require calibration procedures such as temperature modulation and warm-up alignmen, policy-oriented simulations must expose these architectural and parameter choices. Structured documentation of representational assumptions, intervention encoding, and aggregation strategies is necessary to prevent opaque meta-systems from shaping governance decisions.

Conceptually, this work advances three shifts in how LLM-agent simulations are positioned within governance research. First, it reframes simulation from prediction to conditional exploration. Second, it elevates robustness under perturbation over single-point accuracy. Third, it moves beyond surface-level explainability toward deeper auditability of epistemic foundations. By embedding robustness and transparency into the design of policy-oriented simulations, this framework aims to ensure that LLM-agent systems augment policy reasoning without masking the uncertainty and contingency that characterize high-stakes AI governance.

References

[1] Steve Bankes. 1993. Exploratory modeling for policy analysis. *Operations research* 41, 3 (1993), 435–449.

[2] Emily M Bender and Batya Friedman. 2018. Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics* 6 (2018), 587–604.

[3] Dmitri Blueschke, Viktoria Blueschke-Nikolaeva, and Reinhard Neck. 2024. Sensitivity analysis for a dynamic macroeconomic policy game in a monetary union. *Central European Journal of Operations Research* 32, 2 (2024), 507–520.

[4] Stephan Brunow, Susanna Burkert, Alina Chircu, Christian Czarnecki, Jana Erthel, I-Sah Hsieh, Arpit Jain, Stefan Latuski, Joanna Pomaskow, Kanav Sharma, et al. 2026. AI and Stress Testing for a Digital Twin of Society. In *Regenerative Zukünfte und künstliche Intelligenz: Band 3: PROFIT*. Springer, 419–440.

[5] Nicholas Diakopoulos. 2025. Prospective algorithmic accountability and the role of the news media. In *Computer Ethics Across Disciplines: Deborah G. Johnson and Algorithmic Accountability*. Springer, 125–143.

[6] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé Iii, and Kate Crawford. 2021. Datasheets for datasets. *Commun. ACM* 64, 12 (2021), 86–92.

[7] Abe Bohan Hou, Hongru Du, Yichen Wang, Jingyu Zhang, Zixiao Wang, Paul Pu Liang, Daniel Khashabi, Lauren Gardner, and Tianxing He. 2025. Can a society of generative agents simulate human behavior and inform public health policy? A case study on vaccine hesitancy. *arXiv preprint arXiv:2503.09639* (2025).

[8] Joshua A Kroll. 2021. Outlining traceability: A principle for operationalizing accountability in computing systems. In *Proceedings of the 2021 ACM Conference on fairness, accountability, and transparency*. 758–771.

[9] Yiming Li and Dacheng Tao. [n. d.]. Position: AI Agents Are Not (Yet) a Panacea for Social Simulation. ([n. d.]).

[10] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. 2019. Model cards for model reporting. In *Proceedings of the conference on fairness, accountability, and transparency*. 220–229.

[11] Naomi Oreskes, Kristin Shrader-Frechette, and Kenneth Belitz. 1994. Verification, Validation, and Confirmation of Numerical Models in the Earth Sciences. *Science* 263, 5147 (1994), 641–646. arXiv:https://www.science.org/doi/pdf/10.1126/science.263.5147.641 doi:10.1126/science.263.5147.641

[12] Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology*. 1–22.

[13] Christian Sandvig, Kevin Hamilton, Karrie Karahalios, and Cedric Langbort. 2014. Auditing algorithms: Research methods for detecting discrimination on internet platforms. *Data and discrimination: converting critical concerns into productive inquiry* 22, 2014 (2014), 4349–4357.

[14] Jad Georges Sassine and Hazhir Rahmandad. 2024. How does network structure impact socially reinforced diffusion? *Organization Science* 35, 1 (2024), 52–70.

[15] Julius Schlumberger, David Gold, Valeria Di Fant, Gundula Winter, Mehmet Ümit Taner, and Jan Kwakkel. 2026. A review of tools and resources to support decision-making under deep uncertainty. *Environmental Modelling & Software* (2026), 106900.

[16] Jim Yoon, Heng Wan, Brent Daniel, Vivek Srikrishnan, and David Judi. 2023. Structural model choices regularly overshadow parametric uncertainty in agent-based simulations of household flood risk outcomes. *Computers, Environment and Urban Systems* 103 (2023), 101979. doi:10.1016/j.compenvurbsys.2023.101979

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